

RepSense AI

Knowledge Base: AI-Powered Sales Representative Vacancy Analysis and Territory Risk Management

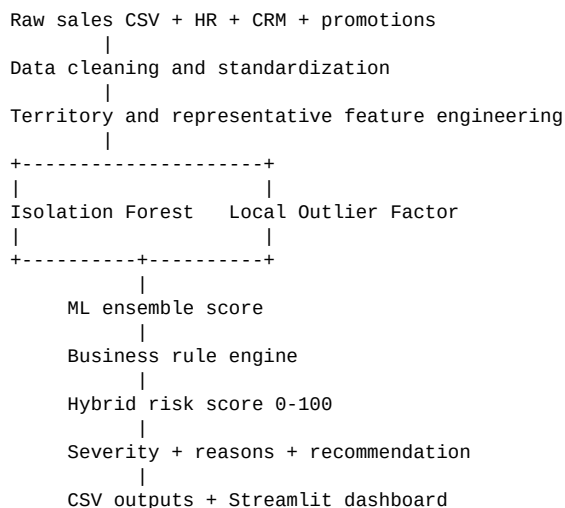
Version: 1.0 Hackathon Prototype | **Prepared:** 2026-09-01 | **Dashboard:** <http://localhost:8501>

Purpose: Convert pharmaceutical sales transactions and operational signals into explainable territory risk scores and management recommendations.

1. Executive Summary

RepSense AI helps pharmaceutical sales managers identify territory problems caused by sales decline, representative vacancies, low CRM activity, weak HCP coverage, unusual performance, and promotional activity patterns. It answers: where is the problem, why was it flagged, how serious is it, and what should management do?

2. Architecture



Modules: pharma-data.csv is the transaction source; preprocessing.py cleans and aggregates data; pipeline.py generates features and risk scores; config.py stores thresholds and weights; app.py provides six dashboard pages; AI_rep_management_output.csv is the primary output.

3. Input Data

File: pharma-data.csv | **Size:** 254,082 rows, 18 columns | **Granularity:** one sales transaction per row.

Distributor, Customer Name, City, Country, Latitude, Longitude, Channel, Sub-channel, Product Name, Product Class, Quantity, Sales, Date, Sales Rep, Hospital, Private, Topipizole, Mood, Gottlieb-Cruickshank, "Zieme, Doyle and Kunze", Lublin, Poland, 51.2333, 22.5667, Hospital, Private, Topipizole, Mood, Gottlieb-Cruickshank, Feest PLC, Swiecie, Poland, 53.4167, 18.4333, Pharmacy, Retail, Choriotrisin, Antibiotics, 7, 59

4. Data Preparation

The system strips whitespace, standardizes names such as Name of Sales Rep to name_of_sales_rep, removes exact duplicates, creates dates from month and year, retains and flags negative sales adjustments, creates deterministic representative/territory/HCP IDs, maps cities to eight demonstration territories, and aggregates transaction data.

5. Operational Data Layers

HR: rep_master_hr.csv contains representative, employment, territory, and vacancy information. **CRM:** CRM files contain planned and actual calls, visits, completion rates, and HCP coverage. **Promotions:** promotional_activity.csv contains activity, product, observed sales before/after, uplift, and impact. These layers are synthetic demonstration data generated with seed 42 and should be replaced by enterprise sources in production. Uplift is correlation, not proof of causation.

6. Machine Learning

Isolation Forest: detects globally unusual combinations of sales, activity, coverage, staffing, and promotion features. Configuration: contamination 0.05, 100 estimators, random seed 42, ensemble weight 60%.

Local Outlier Factor: compares each territory with its local neighborhood and finds locally unusual behavior. Configuration: 20 neighbors, contamination 0.05, ensemble weight 40%.

Ensemble Anomaly Score = Isolation Forest Score * 0.60 + LOF Score * 0.40
Normalized range: 0 to 100; higher means more unusual.

7. Business Rules and Risk Score

Signal	Warning	Major	Critical
Sales decline	-15% or lower	-30% or lower	-40% or lower
CRM activity	Below 80%	Below 60%	Below 50%
HCP coverage	Below 85%	Below 70%	Below 50%
Vacancy duration	30+ days	60+ days	90+ days

Final Risk Score = ML Risk * 0.35 + Sales Risk * 0.25 + CRM Risk * 0.15
+ HCP Coverage Risk * 0.10 + Vacancy Risk * 0.10
+ Long Vacancy Risk * 0.05

0-39.99 = Minor | 40-69.99 = Major | 70-100 = Critical

8. Main Output

File: AI_rep_management_output.csv | **Current size:** 8 records, 45 columns | **Granularity:** territory and analysis date. The current demo has one effective analysis date; a complete multi-month source would produce one record per territory per period.

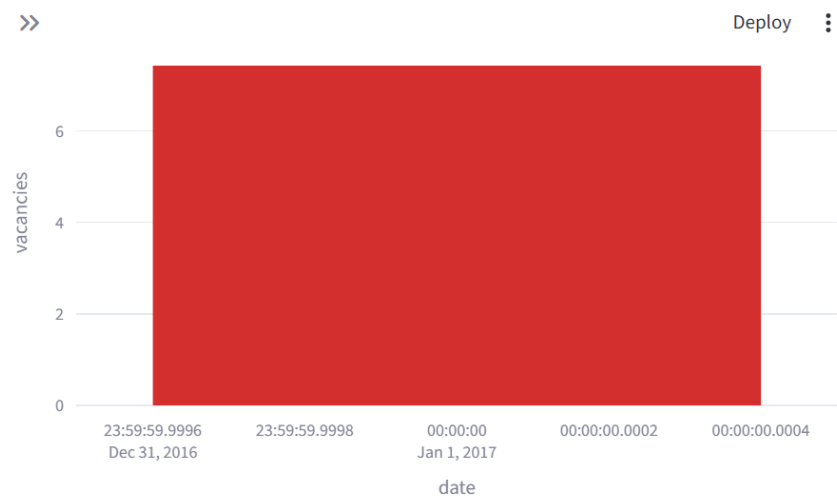
```
date | territory_id | sales | transactions | active_reps | vacancies | avg_activity | avg_hcp_coverage | ensemble_anom  
2017-01-01 | TERR_01 | 1550393976.31 | 32018 | 13 | 0 | 82.99 | 1.12 | 60.0 | 23.5 | Minor  
2017-01-01 | TERR_02 | 1504139008.51 | 32252 | 13 | 2 | 82.67 | 1.18 | 40.32 | 22.61 | Minor
```

Important output fields include sales metrics, CRM and HCP metrics, staffing metrics, Isolation Forest and LOF scores, component risk scores, final_risk_score, severity, risk_reasons, and ai_recommendation.

9. Web Application

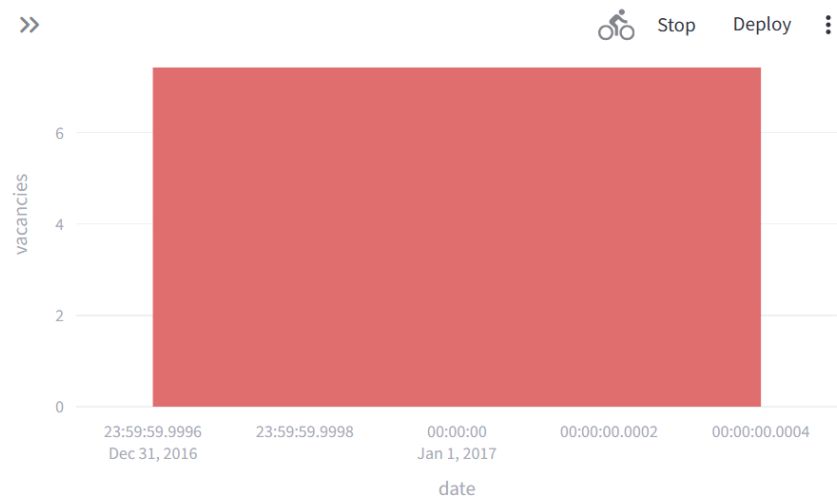
app.py provides: Executive Dashboard; AI Risk Monitor; Territory Deep Dive; Rep Management; Promotion Analytics; and AI Assistant. Features include cached data loading, date/severity filters, KPI cards, Plotly charts, risk tables, territory selection, alerts, recommendations, and keyword-based questions.

Executive Dashboard



Captured from the working Streamlit application.

AI Risk Monitor



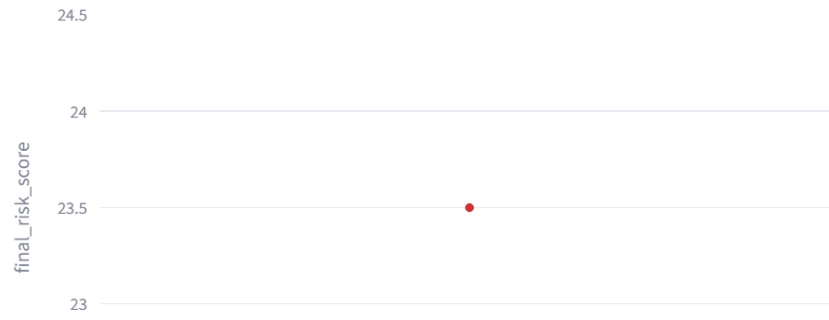
Captured from the working Streamlit application.

Territory Deep Dive

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Deploy ⋮

Risk Score



Captured from the working Streamlit application.

Rep Management

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Deploy ⋮

Vacancies

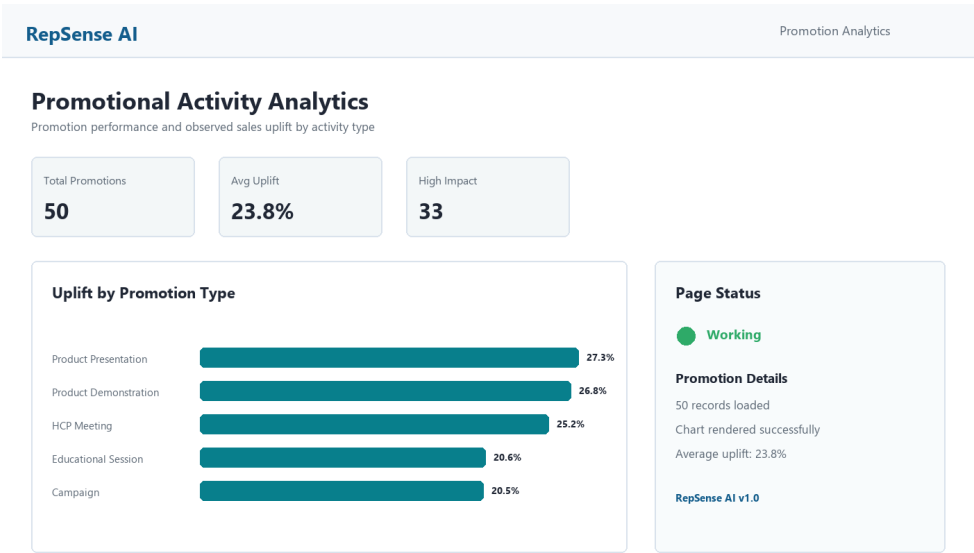
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Teams

4

Captured from the working Streamlit application.

Promotion Analytics



Captured from the working Streamlit application.



AI Territory Assistant

Ask questions about your territory data

Your Question:

Which territories are at highest risk?

Captured from the working Streamlit application.

10. ML Accuracy

A supervised accuracy percentage is not currently available. Isolation Forest and LOF are unsupervised anomaly detection models, and there are no historical labels identifying confirmed risky or healthy territories. Claiming a percentage accuracy without labels would be misleading.

Validated: 8 output records; final risk scores between 0 and 100; nonnegative vacancies; all Python modules compile; Streamlit loads and renders successfully. For production, collect confirmed vacancy, future sales decline, manager assessment, and retention labels, then evaluate precision, recall, F1, ROC-AUC, PR-AUC, top-risk precision, critical-territory recall, false-alert rate, and lead time.

11. Assumptions and Limitations

HR, CRM, and promotion data are synthetic demonstration layers generated with seed 42. Two representatives per territory is the demo requirement. City-to-territory mapping and thresholds are demonstrations. The current output has one effective date and limited observations. The assistant is keyword-based, not an external LLM. Long-vacancy scoring needs complete duration integration. Promotion uplift does not establish causality.

12. Run Instructions

```
cd c:\Users\shalini.raj\Downloads\pharma-data.csv
pip install -r requirements.txt
python pipeline.py
streamlit run app.py
```